

1.12 FREE INDUCTION DECAY (FID)

After an RF pulse tips M_z into the transverse plane, the resulting M_{xy} precesses and induces a decaying voltage in the receive coil.

WHAT IS FID?

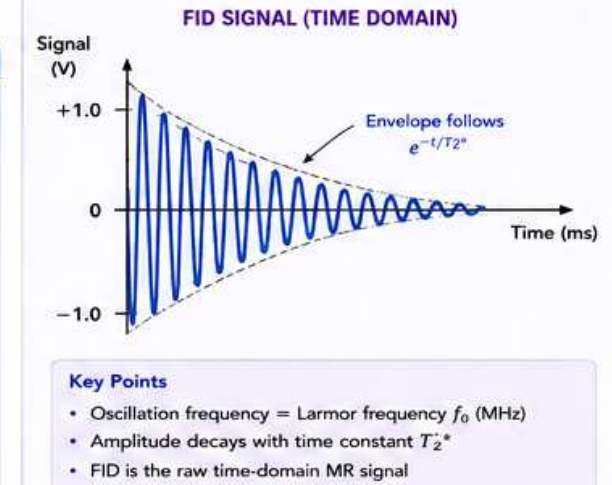
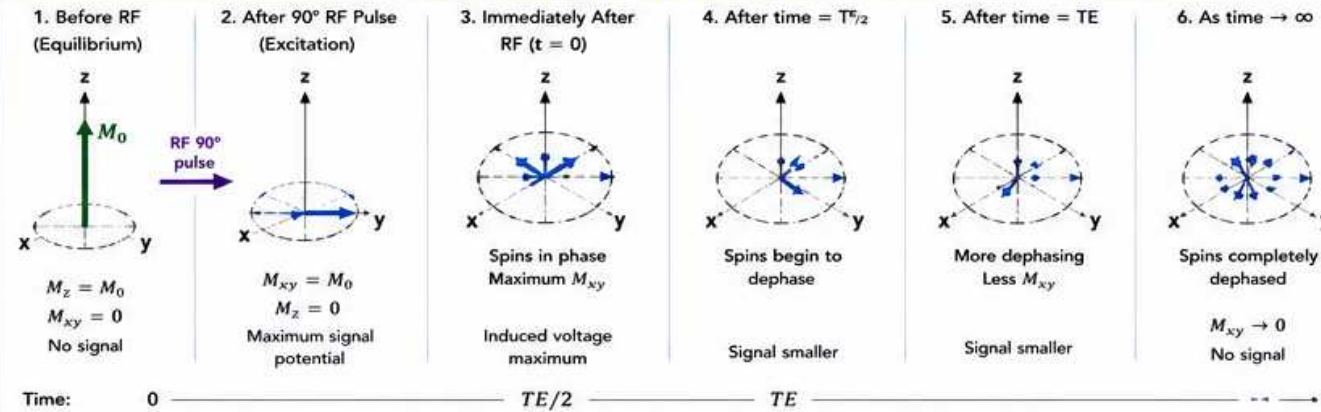
When an RF pulse (excitation) tips the net magnetization into the transverse plane, the spins precess in phase at the Larmor frequency (ω_0).

This rotating transverse magnetization (M_{xy}) induces a voltage in the receive coil (Faraday's Law).

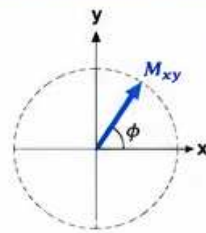
Because spins dephase (T_2^* decay), the signal amplitude decreases exponentially with time.

This is called **Free Induction Decay (FID)**.

FID FORMATION AFTER A 90° RF PULSE

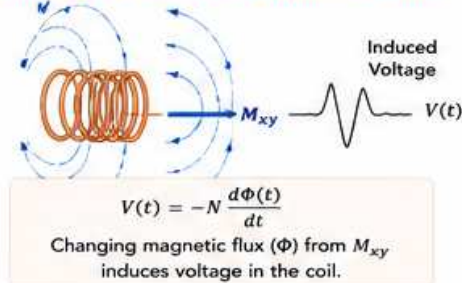


FID PHASE AND FREQUENCY



- M_{xy} rotates in the transverse plane at ω_0 .
- The phase at any time t is $\phi(t) = \omega_0 t + \phi_0$
- Small field inhomogeneities add extra phase ($\rightarrow T_2^*$ decay).
- A perfect, homogeneous magnet would produce no extra dephasing (only T_2).

FID IN THE RECEIVE COIL (FARADAY'S LAW)



FID MATHEMATICS

The FID signal can be modeled as:

$$s(t) = S_0 e^{-t/T_2^*} e^{i(2\pi f_0 t + \phi_0)}$$

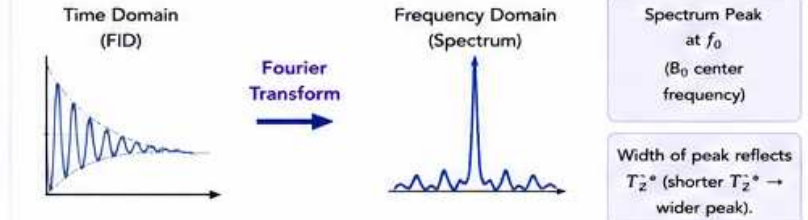
Where:

- S_0 = initial signal amplitude
- T_2^* = apparent transverse relaxation time
- f_0 = Larmor frequency (MHz)
- ϕ_0 = initial phase
- i = imaginary unit

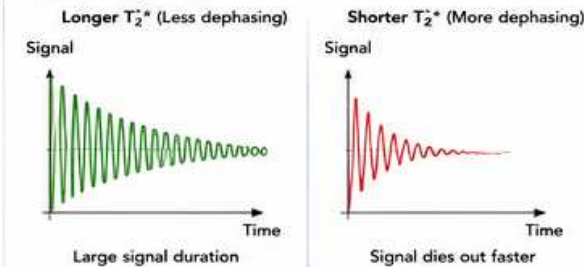
FID exists for all sequences (SE, GRE, IR, etc.).

FID TO FREQUENCY DOMAIN (SPECTRUM)

Applying Fourier Transform to the FID converts the time-domain signal into the frequency-domain spectrum.



EFFECT OF T_2^* ON FID



TYPICAL T_2^* VALUES AT 1.5T (approx.)

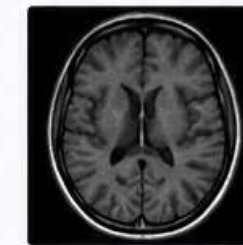
Tissue	T_2^* (ms)
White Matter	40 – 60
Gray Matter	35 – 50
Skeletal Muscle	20 – 30
Liver	15 – 25
Fat	10 – 20
CSF	50 – 100
Myocardium	20 – 30
Iron (hemosiderin)	< 5

Values vary with field strength, sequence, TE, and vendor.

CLINICAL RELEVANCE

- FID exists in all MRI sequences. It is the basis of the MR signal.
- T_2^* effects are important in GRE sequences.
- Used to detect susceptibility differences:
 - Iron, calcium, hemorrhage, microbleeds
 - Deoxygenated blood (BOLD fMRI)
 - Metal artifacts
- Short T_2^* $\rightarrow$ rapid signal loss $\rightarrow$ dark areas on GRE images.
- Foundation for k-space sampling—each k-space line is an FID.

EXAMPLE: GRE IMAGE (Sensitive to T_2^*)



Susceptibility causes signal loss (very short T_2^*).

KEY TAKEAWAY

A 90° RF pulse creates transverse magnetization (M_{xy}). Spins precess in phase and induce a voltage in the coil $\rightarrow$ this is the FID.

The FID oscillates at the Larmor frequency (f_0) and decays with time constant T_2^* .

FID in time-domain $\leftrightarrow$ Spectrum in frequency-domain.

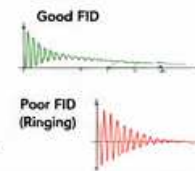
FID is the raw signal that forms the basis of all MRI imaging.

ENGINEERING INSIGHT

- FID is extremely sensitive to:
 - System stability, shimming, gradients
 - Eddy currents and vibrations
- Monitor FID during service:
 - Check for noise, baseline drift
 - Verify preamp, cable, and coil integrity

FIELD SERVICE TIP

- Use vendor service programs to view FID.
- Look for: symmetric decay, no spikes, minimal baseline oscillations.
- Poor shim $\rightarrow$ faster decay (shorter T_2^*).
- Gradient/eddy issues $\rightarrow$ ringing or distortions.



COMMON MISCONCEPTION

"A stronger magnet always gives a stronger signal."

Reality: Higher field increases S_0 , but also increases susceptibility $\rightarrow$ shorter T_2^* $\rightarrow$ faster decay in some tissues.

REAL-WORLD ANALOGY

Think of a bell (spins) struck once (RF pulse). It rings (FID) and gradually gets quieter due to friction and environment (dephasing).

